

Institute of Actuaries of India

Subject ST8 – General Insurance: Pricing

April 2016 Examination

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Obstacles:**

Past data of Injury of other cricketers might not be readily available.

Even if data is available, it is unlikely to have large volume and quality of data might be an issue.

The data might not be relevant as in the past matches played and training facilities were limited. So the trends might have changed.

The data would be required to be divided into various homogeneous groups. E.g. based on characteristics of cricketer like age, drinking habit, etc. which may further reduce the data in each cohort.

[2]

ii) Rating factors:

Per Day Sum Insured

Spread of earning over the year

Type of matches played

Bowler vs. batsman vs. all-rounder

Amount of travel

Health factors – Heart rate, blood pressure, etc.

Diseases covered and exclusions

Age

Gender

Pre-existing diseases

Smoking & drinking habits

Fitness activities

[3]

[5 Marks]

Solution 2:

Get the historic policy and claims aviation data & validate. Check if the fleet of the airlines in historic data matches with the fleet of the airline company which has approached.

The data would be required to split into hull insurance and liability insurance (passenger and public). Each of the component would be modeled based on the below mentioned steps.

De-trend the data and apply IBNYR claim numbers. Estimate the frequency for each prior year.

Fit an appropriate distribution after removing any outlier e.g. Poisson and check the goodness of fit.

Estimate the frequency for middle of exposure period based on the past trend.

Develop individual losses to ultimate using % of IBNR or for individual claim use development factor.

Remove any large claims (and even nil claims) from the data.

Check the severity trend (need for inflation), de-trend it and fit a distribution. Eg Gamma and check the goodness of fit.

Use it to estimate the severity for middle of exposure period allowing for any trend and future inflation.

Add the large loss loading based on the large losses in past data allowing for any trend and inflation.

Use simulation methods to get the combined burning cost distribution for hull and liability after allowing for correlation between them.

Obtain the 95th percentile and mean of the distribution. The mean would be the required risk premium.

[6 Marks]

Solution 3:

i)

Calculate the net commission rate as : $(10\%) - (20\% * 20\%) = 6\%$

Vehicle Age (A)	Depreciation (B)	Claim Severity (C=B*20,000)	Burning Cost (D=C*.3)	Gross Prm (E=D/(1-(10%+5%+6%)))
0-1	10%	2000	600	759
1-2	20%	4000	1200	1,519
2-3	30%	6000	1800	2,278
3-4	40%	8000	2400	3,038

>4	50%	10000	3000	3,797
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Assumptions:

Claim severity consists of labour and part cost. Considering depreciation on severity of Rs.20,000 applies it to both, labour and part cost.

Frequency is likely to change because of benefit including depreciation.

[4]

ii)

- Claim payment including depreciation would have increased the benefit to policyholder so he stops thinking about the premium loss due to NCB for small claims.
Check: Divide the claim into bands and see the increase in frequency in smaller claim bands.
- His driving behavior might be altered as he knows he will get Claim payment before depreciation.
Check: For the renewal customers, see if the frequency of claims has increased for those who have taken this cover.
- The customers who drive long distance/rash might have been more interested to take the add-on as compared with other customers and hence negative selection in portfolio resulting in higher frequency.
Check: The odometer reading/analysis from output of telematics device might be useful to capture the anti-selection.
- Fraudulent activities might have increased due to higher benefit.
Check: Does the score based on fraud indicators show an increase.

[4]

[8 Marks]

Solution 4:

i)

Reinsurance X is said to inure to the benefit of another reinsurance arrangement Y if X acts before Y, and Y acts on the net retained amount after X has been applied.

[1]

ii)

(a) Insured Value	(b) Risk Premium	(c) Retention as % of Insured Value = 100,000/(a)	(d) Retention + Limit as % of Insured Value = 300,000/(a)	(e) Exposure Factor = P(d) – P(c), where P is the given exposure curve	(f) Risk Premium for XoL Treaty = (b)*(e)
25,000	250,100	400%	1200%	0.00%	0
50,000	167,000	200%	600%	0.00%	0
100,000	324,300	100%	300%	3.00%	9,729
250,000	125,750	40%	120%	40.00%	50,300
333,000	51,000	30%	90%	43.00%	21,930
500,000	64,000	20%	60%	37.00%	23,680

Risk Premium for XoL Treaty = [Total in (f)] = 105,639.

[4]

iii) We would need to re-state the subject risk premium net of the insuring reinsurance treaty. For the insured value of 333,000, a retained line of 300,000 would mean that the net risk premium is 300,000 divided by 333,000 = 90.1% of total risk premium (since proportional treaty) = 90.1% * 51,000 = 45,946

For the insured value of 500,000, a retained line of 300,000 would mean that the net risk premium is 300,000 divided by 500,000 = 60% of the total risk premium (since proportional treaty) = 60%*64,000 = 38,400.

Then we set the new net insured values for these original higher limits to 300,000 and apply the given exposure curve with respect to these adjusted insured values.

(a) Insured Value (after Surplus Treaty)	(b) Risk Premium (After Surplus Treaty)	(c) Retention as % of Insured Value = 100,000/(a)	(d) Retention + Limit as % of Insured Value = 200,000/(a)	(e) Exposure Factor = P(d) – P(c), where P is the given exposure curve	(f) Risk Premium for XoL Treaty = (b)*(e)
25,000	250,100	400%	1200%	0.00%	0
50,000	167,000	200%	600%	0.00%	0
100,000	324,300	100%	300%	3.00%	9,729
200,000	125,750	40%	120%	40.00%	50,300
300,000	45,946	33%	100%	44.00%	20,216

300,000	38,400	33%	100%	44.00%	16,896
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Revised Risk Premium for XoL Treaty = [Total in (f)] = 97,141

[3]

[8 Marks]

Solution 5:

i)

Quality of past data: Values of the rating factors should be range bound and the amounts should match with the company's financial statements.

Volume of data: More years help to reduce random fluctuations. If the company is selling huge number of policies then there is a lesser requirement on number of past years.

Change in policy terms and conditions: If the policy terms and conditions have changed, like certain exclusions are now covered, the data will be less appropriate.

Removal of year with an extreme event: If there is an epidemic in any year because of which the frequency has increased sharply (and maybe confined to certain region) and the disease can't be separated out, the year should be removed.

Changes in claims trend from past: Claims environment may have changed, eg, there may be newer diseases, expensive but better treatments, etc. In such case, past data is less reliable.

Undeveloped recent year: There is more uncertainty than a complete year. IBNR is usually applied as a % loading to the base rate while it could be more relevant to certain segment. But retail health is short tailed, so it should not be much of an issue.

[3]

ii)

Chi square

Two nested models can be compared using chi square test for the change in scaled deviance. Two models can be built keeping Sum Insured in one model and excluding it from second.

If the addition of sum insured is statistically significant based on chi square value, addition of this rating factor improves the fit of the model.

F-statistic:

In cases where the scale parameter for the model is not known, its estimator is distributed as a chi square distribution, and the ratio of two chi square distributions gives the F distribution.

Similar to above, if the addition of location factor is statistically significant based on F-statistic, addition of sum insured band improves the fit of the model.

AIC:

If the models are not nested, the AIC can be used. The AIC looks at the trade-off of the likelihood of a model against the number of parameters: the lower the AIC, the better the fit.

For example, if two models fit the data equally well in terms of the log-likelihood, then the model with the fewer parameters is the more parsimonious (and therefore "better").

Thus, if addition of sum insured band lowers the AIC, the fit has improved.

Standard error

The error ranges of the relativities should be distinct (they don't overlap too much) at ± 2 standard, indicating that the response from the data underlying each level has a significantly different relativity value.

Time test

The time consistency check is performed by interacting sum insured band with a time-related factor. If the trend observed for various time bands are consistent, factor is said to be time consistent.

Consistency check with other factors

The trends in interaction of the location factor with other factors should also be consistent.

[5]

- iii) Smoothing would result in grouping of parameters. The lowering of parameters would likely reduce the standard error and improve the predictability of the model. This improvement would be given by AIC statistic.

[1]

[9 Marks]

Solution 6:

i)

- I. Long tail pattern so lot of dependence on IBNR. IBNR is hard to estimate due to factors like out of court settlement, high inflation, change in case reserving philosophy, etc. Also, estimating IBNR at granular level may be difficult.
- II. Not many companies will have many years of data and the freq is quite less, so little claims data. This holds true especially for the smaller or startup companies. The low volume will reduce the certainty of prediction. Not all companies would have been writing all vehicle segments.
- III. The frequency trend has changed over time (it has decreased). Also it varies by geography and vehicle segment. This would further increase the uncertainty in prediction.
- IV. Severity depends on court awards, inflation, other economic factors. So, the future may not follow the past and would lead to additional uncertainty.
- V. Since all companies will file at same time, so no idea about competitive rates which might make company loose/increase motor business.
- VI. Many vehicle categories each having its own characteristics (GCV, PCV, Private car, Two Wheeler, etc). So, the data may be sub-divided to build different models.
- VII. There may be external influence on the price which may prevent sharp rise in price even if the model suggests so (It is a compulsory product to have for any vehicle on road)
- VIII. Companies can use the industry database but would need to adjust it for their own experience and business mix. There may be outliers in the database which need to be taken care of before using the results.
- IX. Orphan claims (claims where insurer is unknown) may further add volatility to frequency & severity trends as future trend of these claims is hard to predict.

[7]

- ii) Court awards could vary by region due to difference in cost of living or difference in assumptions like future wage inflation.

Tendency to fight a legal battle – Human behavior of a society

Presence of companies who urge to fight a legal battle for some fees

Fraud in some locations

Wage gap between cities or rural/urban

Infrastructure – Bad infra may result in large number of accidents and high TP

Traffic density – Slow moving traffic may result in lesser accidents

Legal and medical cost may vary from region to region.

[4]

[11 Marks]

Solution 7:

i)

Reinsurer

Its own risk appetite based on its business objectives, amount of capital, etc.

Its current exposure based on geography. Whether the insurer book will diversify its business or increase the concentration risk.

Its domain experience in Cat reinsurance: Amount of business written, Historic claims experience, rates charged, which perils have been covered.

Is there any support reinsurer has from other reinsurer, i.e., possibility of retrocession, and if yes, at what rates.

Insurers

Distribution of sum Insured: This will help the reinsurer understand small, big and mega risks.

Spread of risk: Based on geography and type of property (House/Shops/offices/factory/etc)

Exposure details divided into various cat zones for various cats like Earthquake, Flood, etc.

Contract details

Details about the layer covered under the cat contract and other RI arrangements taken by the insurer.

Loss details at various percentiles based on OEP/AEP curves constructed from the reinsurers' cat model.

Prevailing market price will help the reinsurer understand the risk vs return.

[5]

ii) a) Future underwriting strategy**Individual policy details which includes:**

Type of property: Domestic/Commercial: shop/office/factory/etc.

Location of the property: GPS coordinates/ Pin code

Sum Insured (/PML)

Limits and deductibles

Other RI contracts before which the Cat triggers

Details about construction: Pillar based/Eq resistant model, floors

Age and size of the property

Individual cat claim details of any previous claims which include:

Policy reference

Claim payment details (Gross and RI)

Peril of loss: Earthquake/Flood/etc

Date of loss

[3]

ii) b)

Difference in assumptions of the cat model: The stochastic events, hazard and vulnerability parameters estimated may vary from reinsurer to reinsurer.

Difference in modeling techniques: The level of modeling granularity may differ, simulation techniques may differ, number of simulations may differ, possible error in model, etc.

Difference in Correlation assumptions: The correlation parameters and dependency distribution used to combine various catastrophe models may be different.

[3]

[11 Marks]

Solution 8:

- i) Historically, there has been a limited focus on the quantity and the quality of data that has been collected on commercial portfolios. This can result in limitations around the data and subsequent analyses. The poor data quality in commercial insurance is due to a number of factors which include
- Data within the commercial insurance enterprise is often fragmented — stored in a variety of siloed systems and databases.
 - Commercial insurers may even have different systems by different line of business — one set of systems for small commercial, another set for workers comp, a third set for mid-market, and so on. This brings into data inconsistency.
 - Commercial insurers tend to run on legacy systems which may not have data validation rules resulting in erroneous or missing entries.
 - Most insurers use a variety of systems for underwriting, policy administration, claims — these were probably developed at different points in time and have been integrated via patches or manual methods.
 - A large proportion of Commercial insurance policies are sold through brokers who aim to minimize their administrative cost and any inconvenience for their end client — they tend to push back on the collection of any additional fields which may be required for pricing
 - As any investment in analytics and management information has not been a priority for Commercial insurance, there has been less incentive for the insurers to have had a focus on data quality
 - A long lag time between implemented improvements in data collection and the ability to see clear business outcomes, resulting in a lack of urgency in making any improvements
- [5]
- ii) Systems for monitoring business typically include a data capture process, calculations and/or manipulations on the data, and a process for reporting the results. Desirable features for such a system are set out below.

Tailored output

Output should generally be concise (“information not data”), and tailored to the strategic goals of the organization. Output should aid decision making.

Accuracy

- Data used should be reliable and validated.
- Calculations should take into account all the key drivers.
- Results should be validated (for example, historical premium rate changes could be compared with changes in emerging loss ratios). This is part of the actuarial control cycle.

Ease of use

- Data should be easy to collect.
- Calculations should not be overly complex.
- System should be documented, extendable and low maintenance.
- Results should be clear and easy for users to interpret.

Consistent

- Output should be consistent over time. If, for example, methods for calculating rates change, then indices should be restated.
- Inputs should be consistent with other data sources. If two quantities are the same (for example, premium) then it should be the same regardless of data source. Otherwise the user will lose confidence.
- Outputs should be consistent with other analyses (for example, results should be shown split into the same business segments).

Timely production of results

The delay between the data cut-off date and the production of results should be kept to a minimum.

[6]

[11 Marks]

Solution 9:

- i) In order to ensure accuracy in ratemaking, especially when measuring trend, it is very important to know the history of the block of business under consideration. I would like to pay attention to the following areas while conducting the rate adequacy study:
- Data Integrity** – Whether all data items have been entered and maintained in a consistent manner throughout the history of the database? Are changes in coverage reflected in changes in plan/coverage codes?

Vehicle Coverage - What changes in coverage options, term/mileage availability, deductible options, and vehicle eligibility categories have taken place over the years? Which are the key time points in past when such changes were introduced?

Benefits - Have ancillary benefit packages (rental reimbursement, substitute transportation, towing etc.) changed in composition or in the extent/nature of the benefits provided?

Claims Adjustment Policies - What changes have been made in the interpretations of coverages, conditions and exclusions over the years? Which are the key time points in past when such changes were introduced?

Rating Structure – What type of changes have taken place in past in the method of rating vehicles? Was there any change in vehicle classification?

Vehicle Mix - Has the mix of makes, models, equipment changed enough to affect trends in composite loss development patterns or ultimate losses? Has the geographic mix of business changed over the years under study?

[5]

ii) The calculations are shown in the below table:

Year	Cumulative Miles	Miles within Manufacturer's warranty	Cumulative exposure (Miles)	Increment (Miles)	Claims trend	Mileage factor	Adjusted Exposed Miles	Percent Exposure
1	15,000	15,000	0	0	1.00	1.00	0	0%
2	30,000	30,000	0	0	1.03	1.04	0	0%
3	45,000	36,000	9,000	9,000	1.06	1.08	10,327	23%
4	60,000	36,000	24,000	15,000	1.09	1.12	18,438	41%
5	75,000	36,000	36,000	12,000	1.13	1.17	15,800	35%
6	90,000	36,000	36,000	0	1.16	1.22	0	0%

So in this example, we could assume that the earnings pattern should be 23% in Year 3, 41% in Year 4, and 35% in Year 5. Nothing would earn in Year 6 due to contract expiring due to miles. Years 1 and 2 would also earn nothing due to the manufacturer's warranty.

It is assumed that the claim frequency will increase 4% year on year and not directly depending upon the miles driven.

[5]

- iii) The key issue in this approach is that the specific driving habit of a contract holder is unknown. In this example, we have assumed the vehicle driven on an average 15,000 miles per year and hence the driver's mileage exceeds the maximum covered by the extended warranty in Year 5. That may be true for average driver on the book, but one could expect some earnings in the 6th year for drivers who are driving fewer miles than the average for the book. [1]

- iv) For the initial mix of business, the loss cost per contract is calculated in the table below.

Original mix of risks

Appliance	Number of contracts	Loss cost per contract (in INR)	Total loss cost (in INR)
Television	1,200	7,500	9,000,000
Refrigerator	2,000	10,000	20,000,000
Microwave	600	5,000	3,000,000
Total	3,800	8,421	32,000,000

Loss cost per contract = $32,000,000 / 3,800 = 8,421$

Expected loss ratio = 67%

Gross rate per contract = $8,421 / 67\% = 12,569$ (in INR)

Similar calculation is done after including the new account sold by the TPA.

New mix of risks

Appliance	Number of contracts	Loss cost per contract (in INR)	Total loss cost (in INR)
Television	1,200	7,500	9,000,000
Refrigerator	3,000	10,000	30,000,000
Microwave	600	5,000	3,000,000
Total	4,800	8,750	42,000,000

Loss cost per contract = $42,000,000 / 4,800 = 8,750$

Expected loss ratio = 67%

Gross rate per contract = $8,750 / 67\% = 13,060$ (in INR)

Due to this changes in the mix of risks, loss cost per contract changed making the single rate of 12,569 INR inadequate as the new rate should be 13,060 INR.

It is not practical to modify rates every time a new account comes on book. The Company, in this case, is facing this problem as it decided one single premium for the entire book, whereas it would have been better to charge different rates for different classes to minimize the mix change problem.

[4]

[15 Marks]

Solution 10:

i) Crop insurance is usually offered on an indexed basis or on an indemnity basis.

In index basis, the claims are a function of a defined index chosen to be a good proxy for incurred crop loss. Claims are triggered when there is a deviation on the set weather indices and not by actual yields. It pays a fixed amount for each level of variance in the weather indices. The sum Insured and claim payable at various trigger level of weather parameter are pre-defined.

On indemnity basis, the claims are based on loss of yields due to the perils covered. Losses arising from yields falling below the average local yields are paid by the insurer. The sums insured can also be based on the loan amount.

[4]

ii) Some of the important factors influencing crop insurance product design are as follows:

- Climate and weather conditions
- Type of crop
- Need of farmers in terms of insurance coverage
- Farming expertise
- Importance of the crop in the national economy
- Political stability, system and leadership
- Prevalent Disaster and event management systems

- Banks, financial institutions and insurers
- Agriculture establishments, crop monitoring systems and controls
- Cultural, social and economic set up of the country – awareness among farmers
- Administrative systems
- Network, data management, weather forecasting and information systems
- Cross subsidies with other products

[4]

iii) a) Spatial means “relating to space”. The spatial smoothing techniques are based on the assumption that, in the absence of credible experience, it is reasonable to assume that neighbouring areas have similar risk levels. [1]

b) Two main forms of spatial smoothing are:

- Distance-based smoothing
- Adjacency-based smoothing

Distance-based smoothing incorporates information about nearby location codes based on the distance between the location codes: the further away a location code, the less influence (or weight) is given to its experience.

Adjacency-based smoothing incorporates information about directly neighboring location codes. Each location code is influenced by its direct neighbors, each of which is in turn influenced by its direct neighbors; distributional assumptions or prior knowledge of the claims processes can be incorporated in the technique.

Distance-based smoothing methods are often employed for weather-related perils

[4]

iv) Basis risk is present where the coverage is dependent on some index.

Weather-index insurance, which does not physically verify the crop losses caused, but estimates them on the basis of one or more triggers (rainfall and temperature being commonly used to compute the trigger level) is prone to basis risk. On the other hand, area-yield loss index insurance does not set a trigger dependent on weather data, but the insurer or a nominated entity physically verifies the yield losses to compensate the insured.

In order to handle the basis risk, the product should minimize the difference between insurance payouts and losses suffered by farmers.

Weather-index insurance has advantages of: a) lower moral hazard i.e. a farmer's action of making the claim event happen or report a higher claim is reduced, and b) lower anti-selection risk i.e. a farmer's electing to purchase an insurance policy knowing well that the event is likely to affect the farm output/ livestock losses.

[3]

[16 Marks]
